

Section 5: Chemistry in society

- a) Extraction and uses of metals
- b) Crude oil
- c) Synthetic polymers
- d) The manufacture of some important chemicals

- a) Extraction and uses of metals

Students will be assessed on their ability to:

- 5.1 explain how the methods of extraction of the metals in this section are related to their positions in the reactivity series
- 5.2 describe and explain the extraction of aluminium from purified aluminium oxide by electrolysis, including
 - i the use of molten cryolite as a solvent and to decrease the required operating temperature
 - ii the need to replace the positive electrodes
 - iii the cost of the electricity as a major factor
- 5.3 write ionic half-equations for the reactions at the electrodes in aluminium extraction
- 5.4 describe and explain the main reactions involved in the extraction of iron from iron ore (haematite), using coke, limestone and air in a blast furnace
- 5.5 explain the uses of aluminium and iron, in terms of their properties

- b) Crude oil

Students will be assessed on their ability to:

- 5.6 recall that crude oil is a mixture of hydrocarbons
- 5.7 describe how the industrial process of fractional distillation separates crude oil into fractions
- 5.8 recall the names and uses of the main fractions obtained from crude oil: refinery gases, gasoline, kerosene, diesel, fuel oil and bitumen
- 5.9 describe the trend in boiling point and viscosity of the main fractions
- 5.10 recall that incomplete combustion of fuels may produce carbon monoxide and explain that carbon monoxide is poisonous because it reduces the capacity of the blood to carry oxygen
- 5.11 recall that, in car engines, the temperature reached is high enough to allow nitrogen and oxygen from air to react, forming nitrogen oxides
- 5.12 recall that fractional distillation of crude oil produces more long-chain hydrocarbons than can be used directly and fewer short-chain hydrocarbons than required
- 5.13 describe how long-chain alkanes are converted to alkenes and shorter-chain alkanes by catalytic cracking, using silica or alumina as the catalyst and a temperature in the range of 600–700°C

c) Synthetic polymers

Students will be assessed on their ability to:

- 5.14 recall that an addition polymer is formed by joining up many small molecules called monomers
- 5.15 draw the repeat unit of addition polymers, including poly(ethene), poly(propene) and poly(chloroethene)
- 5.16 deduce the structure of a monomer from the repeat unit of an addition polymer
- 5.17 recall that nylon is a condensation polymer**
- 5.18 understand that the formation of a condensation polymer is accompanied by the release of a small molecule such as water or hydrogen chloride**
- 5.19 recall the types of monomers used in the manufacture of nylon**
- 5.20 draw the structure of nylon in block diagram format**

d) The industrial manufacture of chemicals

Students will be assessed on their ability to:

- 5.21 recall that nitrogen from air, and hydrogen from natural gas or the cracking of hydrocarbons, are used in the manufacture of ammonia
- 5.22 describe the manufacture of ammonia by the Haber process, including the essential conditions:
 - i a temperature of about 450°C
 - ii a pressure of about 200 atmospheres
 - iii an iron catalyst
- 5.23 understand how the cooling of the reaction mixture liquefies the ammonia produced and allows the unused hydrogen and nitrogen to be recirculated
- 5.24 recall the use of ammonia in the manufacture of nitric acid and fertilisers
- 5.25 recall the raw materials used in the manufacture of sulphuric acid**
- 5.26 describe the manufacture of sulphuric acid by the contact process, including the essential conditions:**
 - i a temperature of about 450 °C**
 - ii a pressure of about 2 atmospheres**
 - iii a vanadium(V) oxide catalyst**
- 5.27 recall the use of sulphuric acid in the manufacture of detergents, fertilisers and paints**
- 5.28 describe the manufacture of sodium hydroxide and chlorine by the electrolysis of concentrated sodium chloride solution (brine) in a diaphragm cell**
- 5.29 write ionic half-equations for the reactions at the electrodes in the diaphragm cell**
- 5.30 recall important uses of sodium hydroxide, including the manufacture of bleach, paper and soap; and of chlorine, including sterilising water supplies and in the manufacture of bleach and hydrochloric acid**